

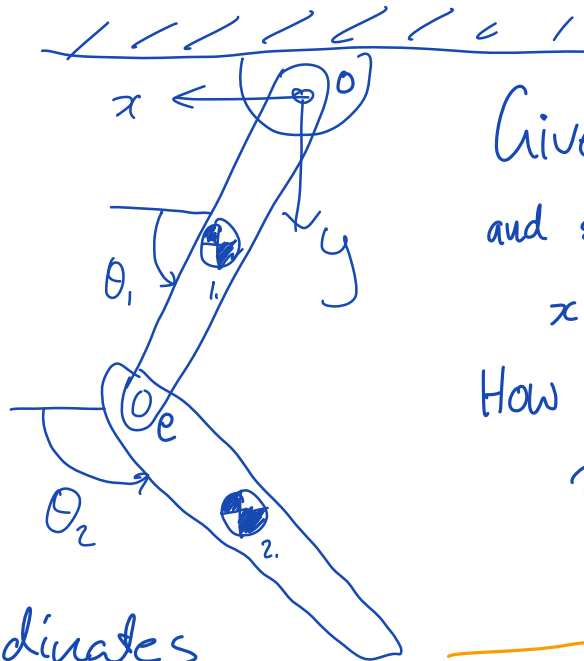
Introduction to Constraints

- o Double Pend.
 - o Simple Cart
- Maximal Coords
 - Minimal Coords
 - Lagrange Eqⁿs

HW5 due
Today 6pm
Prelim 1 next
week
10/6-7.

Complicated Example!

Double Pendulum



Given m_1, m_2, I_1, I_2 etc.
and some initial conditions

$$x_{10}, y_{10}, \dot{x}_{10}, \dot{y}_{10}, \dots$$

How can we find EoM?

3 Methods

1. Maximal Coordinates

'PAE' Approach

Consider separate bodies with DoFs and apply constraints

eg. 6 DoF

4 unknown Reactions

6 N-E eqⁿs

10 eqⁿs

$$[\ddot{x}, \ddot{y}, \ddot{\theta}]_1, [\ddot{x}, \ddot{y}, \ddot{\theta}]_2 + [R_{ox}, R_{oy}, R_{ex}, R_{ey}] \Rightarrow \begin{matrix} 6 \text{ N-E eq}^n\text{s} \\ + \\ 4 \text{ constraint eq}^n\text{s} \end{matrix} \Rightarrow \begin{matrix} 10 \text{ eq}^n\text{s} \\ \Rightarrow \text{in } 10 \text{ unknowns} \end{matrix}$$

Computationally Direct

2. Minimal Coordinates "Reduced"

eg. 2 variables of interest $\ddot{\theta}_1, \ddot{\theta}_2$

Using N-E tricks, solve AMP, LMB, conservation of energy

Intuitive, indirect

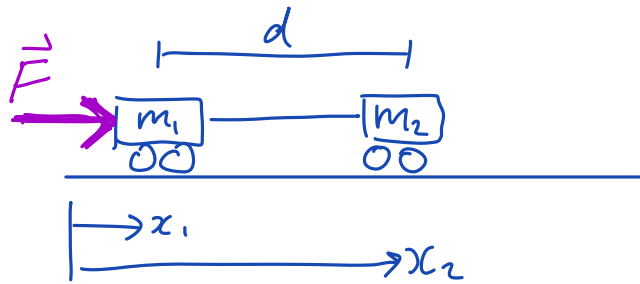
3. Lagrange Equations

Generalized Coordinates q_1, q_2

$$\text{let } \mathcal{L} = E_k - E_p$$

$$\text{Solve } \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i} \right) - \frac{\partial \mathcal{L}}{\partial q_i} = Q$$

!?

Simple Example in 1D.

Find EoM

$$\ddot{x}_1, \ddot{x}_2$$

2. Minimal CoordinatesObserve that $x_2 = x_1 + d$.

$$\frac{d^2}{dt^2} \Rightarrow \ddot{x}_2 = \ddot{x}_1 = \ddot{x}_a$$

(a) LMB $\Sigma F = ma$

$$F = m_T \ddot{x}_a$$

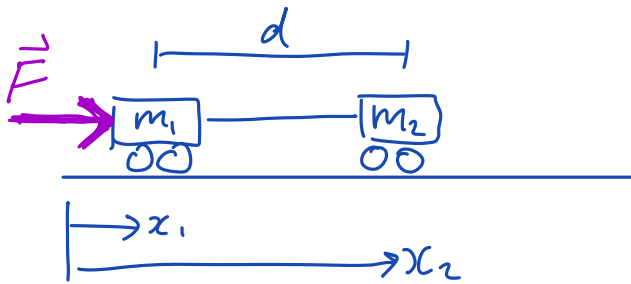
$$\ddot{x}_a = \frac{F}{m_1 + m_2}$$

(b) Energy $P = \dot{E}_k$

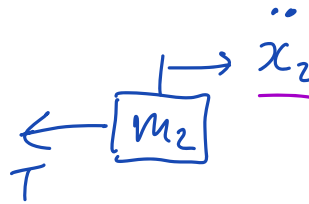
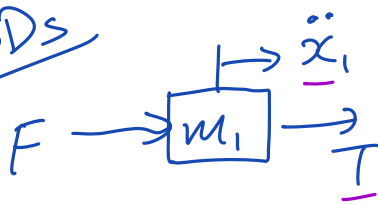
$$\begin{aligned} F \dot{x}_a &= \frac{d}{dt} \left(\frac{1}{2} m_T \dot{x}_a^2 \right) \\ &= \frac{m_T}{2} \left(\dot{x}_a \ddot{x}_a + \ddot{x}_a \dot{x}_a \right) \\ &= m_T \dot{x}_a \ddot{x}_a \end{aligned}$$

$$\therefore \ddot{x}_a = \frac{F}{m_T} = \frac{F}{m_1 + m_2}$$

(assuming $\dot{x}_a \neq 0$)

Simple Example in 1D.

Find EoM

1. Maximal CoordinatesFBDs

3 unknowns
 $\Rightarrow$ 3 eqns.

LMB $\Sigma F = ma$

FBD1: $F + T = m_1 \ddot{x}_1$
 $m_1 \ddot{x}_1 - T = F \quad (1)$

FBD2: $-T = m_2 \ddot{x}_2$
 $m_2 \ddot{x}_2 + T = 0 \quad (2)$

Constraints (kinematics)

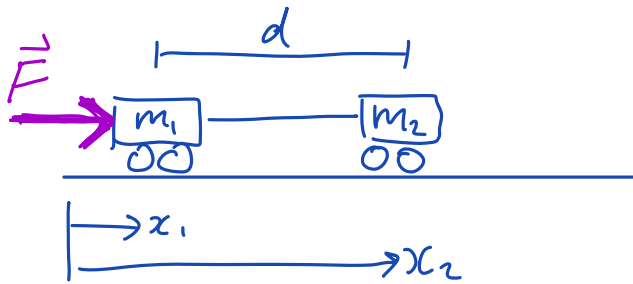
$$x_1 + d = x_2$$

$$\frac{d^2}{dt^2} \Rightarrow \ddot{x}_1 = \ddot{x}_2$$

$$\ddot{x}_1 - \ddot{x}_2 = 0 \quad (3)$$

Solve (1, 2, 3)

$$\begin{bmatrix} m_1 & 0 & -1 \\ 0 & m_2 & 1 \\ 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} \ddot{x}_1 \\ \ddot{x}_2 \\ T \end{bmatrix} = \begin{bmatrix} F \\ 0 \\ 0 \end{bmatrix} \Rightarrow \text{solve this in MATLAB.}$$

Simple Example in 1D.

Find EoM

3. Lagrange Equations

[We will revisit this after
the Fall Break]



